D · Teacher guide, mapping and marking

Distribution and timing

Use the same three-part order; reveal X/Y identities only in the final learning points. The paper group learns from supplied materials without operating the VL. Save first responses before checking the answer copy. One hour is simulated, not classroom waiting time. Use equal total time for both groups; pilot 20–25 minutes before claiming 15–20 is sufficient.

Experiments and science

Main: 2 mL X versus 2 mL Y. Extension 1: two fresh tubes; pretreat with 1 mL X/water and identical shaking, then 1 mL Y each. Equal pretreatment times. Extension 2 uses the same unboiled sequence as baseline, boiling only one juice and cooling to the same temperature. Count one hour after Y addition.

Evidence limits

X is bile; Y is pancreatic juice containing lipase. Emulsification leaves lipid molecules unchanged. Reduced cloudiness / loss of an oil layer are preset appearances, not product tests or measured rates. Explain mechanisms with the final molecular model; glycerol/fatty acids follow the simplified complete-hydrolysis model.

Current marks and legacy records

staged-1: SPS 23 (4 observation, 4 classification, 3 planning, 4 conducting, 4 inference, 4 communication) + reflection 8 = 31. WS 18. Removed control-design field is not applicable, not mandatory. Five inference items earn 0.8 each. Preserve original trial sets, four/six-item inference and totals 32/28 for legacy records.

Manual scores and Excel

Six current manual fields: initial observation 0–2, original prediction/reason 0–2, setup 0–2, fair arrangements 0–2, communication 0–4 and reflection 0–8. Assumptions add automatic 0–1 to planning. Setup requires only the two main tubes. Do not penalise a reasonable prediction for being unsupported; clicks/time are not ability scores.

Saving and versions

Blank means pending, zero is a marked zero, inactive fields need no mark. Formulas reject out-of-range, negative, fractional and text entries. Overall requires reflection submission and all applicable marks. Save a marked workbook separately; re-export does not retrieve manual marks. Preserve IDs, first/final responses, images and snapshots. WS and VL totals are not interchangeable or validated scales.

Worksheet mapping and marking

1. 1. Variables (4): IV = combination of juices; DV = mixture appearance; five CVs worth 0.4 each, wrong-class selection makes CV zero. Same full candidate list as VL.

2. 2. Assumptions (1): time, volumes and shaking; do not select extra volume for X. All correct with no wrong selection earns 1.

3. 3. Main q1 (1): cloudy emulsion.

4. 4. Main qY (1): thinner oil layer, with two layers remaining.

5. 5. Extension 1 q3 (1): increases.

6. 6. Extension 1 limitations (1): appearance alone cannot prove glycerol and fatty acids formed.

7. 7. Extension 2 q2 (1): Y.

8. 8. Final integration (8): use the eight concept points below. No extra observation citation, limits or hypothesis revision required.

Integrated explanation: one mark per concept

1. Bile physically emulsifies large droplets into smaller droplets.

2. Lipid molecules remain unchanged in emulsification.

3. Greater total surface area helps lipase contact lipids.

4. Lipase in pancreatic juice catalyses chemical digestion.

5. Glycerol and fatty acids form in the complete-hydrolysis model.

6. Lipase is a protein.

7. Boiling denatures lipase and stops catalysis.

8. Bile retains emulsifying action after boiling in this model.

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